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Numerical study on the temperature characteristic of material absorptivity and its significance in high-power laser beam welding

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Abstract

The absorptivity of metallic materials plays an important role in high-power laser beam welding. It affects the amount of absorbed laser power leading to the heating and melting of the surfaces to be joined and is highly dependent on the temperature. Nonetheless, this key characteristic is often ignored in numerical simulations and an empirical parameter determined by trial-and-error approaches is rather implemented to calibrate the results. In the present work, the temperature dependence of laser absorption is included in a three-dimensional multiphase numerical model considering the coupled fluid flow and heat transfer. The calculated laser absorption is determined by the temperature-dependent material properties, laser characteristics, and incident angle of the laser beam. It is found that the temperature dependence of the laser absorption is crucial for accurately determining the keyhole and weld pool geometries, which is validated by experimental measurements using 304 austenitic steel.

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1. Introduction

With increasing demands and the need for highly resistant components, deep penetration laser beam welding (LBW) has emerged as a crucial joining process in various industrial sectors in recent years. Nowadays, this highly precise welding process is, amongst others, used in the pipeline industry, powerplant, pressure vessel industry, and shipbuilding [1, 2]. The core of this process lies in the utilization of laser radiation with high laser intensities leading to local heating of the welded material up to vaporization, thus forming a vapor cavity called keyhole. Energy absorption into the material occurs through multiple reflections of the laser radiation within the keyhole, which characterizes this welding process [3].

The absorption of laser energy, particularly at high intensities larger than 10^6 W/cm², is a very complex process. It

significantly influences the heat input and, consequently, affects the weld pool dynamics. However, in-situ experimental observations of the absorption process inside the keyhole are considered unfeasible due to the extreme conditions during deep penetration LBW [4].

Nevertheless, in order to gain more insights into the absorption mechanisms, simulation work has been established. Modeling absorption in deep penetration LBW physically plays a significant role and enables a better understanding of the complex mechanisms of laser beam absorption during the process. Typically, laser beam absorption is simulated using an empirical model that requires calibration parameters [5, 6]. Hereby, the importance of other physical factors, such as the wavelength of the laser radiation, as well as other material related properties, are integrated into a single parameter to be calibrated. However, a big challenge occurs in precisely

calibrating this parameter to facilitate realistic simulations. This may also reduce the model reliability as a change in process parameters or material properties may necessitate recalibration of the parameter. A particular focus of this work lies in the modeling absorption and the associated calibration issue, presenting a physics-based model without any use of calibration parameters as a promising state of the art approach [7].

In this work, two different laser absorption models are simulated and compared with the experimental result. The first model is developed based on empirical method, while the second one incorporated the physic behind laser absorption process. It is found that the temperature characteristic of laser absorption is substantial for the underlying heat input distribution during LBW and a realistic weld pool geometry.

2. Experimental methods

To validate the simulation results, bead on plate welding is performed using a Trumpf disk laser system with a maximum output power of 16 kW. In this experiment, the base material was austenitic steel AISI 304. The dimensions of the base material are 200 mm × 60 mm × 10 mm. Prior to the welding process, the workpieces were cleaned by using ethanol. Pure argon gas was used as the shielding gas with a flow rate of 20 l/min. The optical parameters of the laser system and the process parameters are listed in Table 1.

After the welding process, the specimens were cut transversely and then polished for the preparation of the metallurgical observation. To reveal the melt contour, the cut samples were etched with the etchant V2A (composition: 100 ml H₂O, 100 ml HNO₃, and 10 ml HCl). The cross-sectional images were then captured subsequently using an optical microscope.

Table 1. Optical and process parameters used in LBW experiment.

Parameter	Value
Laser spot diameter	0.42 mm
Wavelength	1030 nm
Focal position (from top surface)	-3 mm
Rayleigh length	3.6 mm
Beam quality	4 mm mrad
Laser power	4 kW
Welding speed	1 m/min

3. Mathematical modelling

In this work, a 3D multi-physical Computational Fluid Dynamics (CFD) model has been developed to simulate the formation of the keyhole coupled with heat transfer from laser-material interaction, as well as the resulting weld pool dynamics. Based on the Finite Volume Method (FVM), the two-phase flow state comprising shielding gas layer and workpiece is computed, allowing for the transient evolution of the keyhole to be reproduced. The free surface of the workpiece and its deformation during the LBW process were modeled using the Volume-of-Fluid method (VOF). Simulating additional physical aspects of deep penetration LBW requires coupling the VOF method with the constitutive laws such as

mass balance, momentum balance, and energy balance, leading to increased complexity in numerical solution efforts. In this regard, some assumptions should be made:

- The molten material is a Newtonian, incompressible fluid.
- The fluid flow behaves laminar with low Reynolds number.
- Boussinesq assumption is considered to model the buoyancy force since the inertia difference of the molten material due to temperature change is negligible.
- Influence of metal vapor on the keyhole wall is implemented empirically [8].

Subsequently, the mentioned constitutive laws as well as the momentum and energy acting on the keyhole wall were calculated and solved by the commercial software ANSYS Fluent 2022 R2 to get the temperature, velocity fields, and free surface deformation. The boundary conditions, calculation domain, and further material modeling used in this work are based on the previous studies and are therefore not further discussed [9, 10].

To model the multiple reflection of laser beam inside the keyhole, the ray-tracing method was employed. The whole laser beam was discretized into a bundle of subrays, each possessing its own power and following a specific propagation direction determined by the beam caustic and laser intensity distribution [6]. The beam-matter interaction was calculated using a localized Level-Set algorithm, which reconstructed the free surface of base material through the Level-Set distance function [11]. Two main improvements in ray-tracing method have been added here compared to author's previous work, which facilitate more accurate description of laser beam spatial distribution as well as the amount of absorbed laser energy [12, 13]. The first improvement includes the implementation of a radius correction factor for the calculation area of the laser intensity distribution to capture 99% of the total laser energy. The second one refers to approximation of laser beam caustic by using different converging and diverging regions of the laser beam [11].

In this work, two different absorption models were implemented and analyzed, i.e., an empirical and a physics-based absorption model. In the empirical absorption model, the Fresnel absorption can be simplified by two assumptions, i.e., the metal has a good thermal conductivity, and the laser irradiation belongs to the infrared spectrum [5]. Then, Fresnel's absorption can be formulated as a function of an empirically calibrated parameter ε and the subray incident angle θ relative to the surface normal direction as follow [6]:

$$A = 1 - \frac{1}{2} \left[\frac{1 + (1 - \varepsilon \cos(\theta))^2}{1 + (1 + \varepsilon \cos(\theta))^2} + \frac{\varepsilon^2 - 2\varepsilon \cos(\theta) + 2\cos^2(\theta)}{\varepsilon^2 + 2\varepsilon \cos(\theta) + 2\cos^2(\theta)} \right] \quad (1)$$

The calibrated parameter ε can be calculated empirically as a combination of constant physical properties of metal and laser wavelength. Hence, the correlation between Fresnel absorption and surface temperature are not considered in the empirical model. However, the calculation of ε serves merely as an estimated value, determined by a calibration to experimental result since the keyhole wall is boiling and this calculated value

tends to underestimate the laser absorption to the keyhole wall. A typical calibrated value of 0.25 was used in the literature research for LBW of mild steel [6, 14], thus three values with difference of 0.05 have been simulated for a comparative study: $\varepsilon = 0.2$, $\varepsilon = 0.25$ and $\varepsilon = 0.3$, which are typically used in many studies.

In contrast to the previous model, the physics-based absorption model considers various physical factors, including laser beam properties, surface temperature, base metal composition, temperature dependent electrical conductivity, and the subrays incident angle with respect to the surface normal direction. The Fresnel absorptivity A for unpolarized laser beam holds:

$$A = 1 - \frac{R_s + R_p}{2}, \quad (2)$$

where R_s and R_p are the perpendicular and parallel component of reflection coefficient:

$$R_s = \frac{\alpha^2 + \beta^2 - 2\alpha \cos(\theta) + \cos^2(\theta)}{\alpha^2 + \beta^2 + 2\alpha \cos(\theta) + \cos^2(\theta)}, \quad (3)$$

$$R_p = \left(\frac{\alpha^2 + \beta^2 - 2\alpha \cos(\theta) + \cos^2(\theta)}{\alpha^2 + \beta^2 + 2\alpha \cos(\theta) + \cos^2(\theta)} \right) \times \left(\frac{\alpha^2 + \beta^2 - 2\alpha \sin(\theta) \tan(\theta) + \sin^2(\theta) \tan^2(\theta)}{\alpha^2 + \beta^2 + 2\alpha \sin(\theta) \tan(\theta) + \sin^2(\theta) \tan^2(\theta)} \right). \quad (4)$$

Herein, Eq. (2), (3), and (9) form the foundation for the physics-based absorption model. Parameter α and β can be calculated as follow:

$$\alpha = \left(\frac{\sqrt{(n^2 - k^2 - \sin^2(\theta))^2 + 4n^2k^2}}{2} + \frac{n^2 - k^2 - \sin^2(\theta)}{2} \right)^{\frac{1}{2}}, \quad (5)$$

and

$$\beta = \left(\frac{\sqrt{(n^2 - k^2 - \sin^2(\theta))^2 + 4n^2k^2}}{2} + \frac{n^2 + k^2 + \sin^2(\theta)}{2} \right)^{\frac{1}{2}}, \quad (6)$$

where n and k are the real and imaginary part of complex refractive index $\tilde{n}$. For a non-magnetic material, the complex refractive index $\tilde{n}$ can be correlated directly from complex relative permittivity $\tilde{\varepsilon}$:

$$\tilde{n} = n - ik = \sqrt{\tilde{\varepsilon}}, \quad \tilde{\varepsilon} = \varepsilon_r - i\varepsilon_i, \quad (7)$$

where ε_r and ε_i are the real and imaginary part of the complex relative permittivity. Subsequently, Eq. (9) left can be rewritten based on real and imaginary part which resulted in:

$$n = \left(\frac{\sqrt{\varepsilon_r^2 + \varepsilon_i^2} + \varepsilon_r}{2} \right)^{\frac{1}{2}}, \quad (8)$$

$$k = \left(\frac{\sqrt{\varepsilon_r^2 + \varepsilon_i^2} - \varepsilon_r}{2} \right)^{\frac{1}{2}}. \quad (9)$$

In this work, parameters ε_r and ε_i can be determined as functions of temperature dependent electrical conductivity and material parameters. Fresnel absorption A emerges consequently as a function of surface temperature and subray angle. The absorption process is elucidated by Lorentz-Drude model phenomenologically which describes laser absorption as a damped harmonic oscillation of the base metal electron driven by laser irradiation. Thus, there is no longer a requirement for a calibrated parameter, unlike in the empirical absorption model. Since this model doesn't necessitate any calibration, it is transferable to other materials and different laser as well as welding parameters. The equations needed for this physic-based model are adopted from Yang et al. [7]. The physical properties needed for this physics-based model are also listed in the following Table 2.

Table 2. Physical properties used in physic-based absorption model.

Parameter	Value
Electron density of AISI 304	$1.53974 \times 10^{29} \text{ m}^{-3}$
Electron mass	$9.1093837 \times 10^{-31} \text{ kg}$
Elementary charge	$1.60217663 \times 10^{-19} \text{ C}$
Vacuum permittivity	$8.85418782 \times 10^{-12} \text{ F/m}$
Electrical conductivity	Extrapolated value from [15] $\sigma(T) = (1.29 \times 10^6 \exp(-2.16 \times 10^{-3} T) + 6.98 \times 10^5) \text{ S/m}$

4. Results and discussion

4.1. Absorptivity of empirical and physics-based model

The laser absorption behavior across the surface temperature and the subray angle to the surface normal are depicted in Figure 1 for both empirical and physics-based model. Both plots are resulted from Eq. (1) for empirical model and Eq. (2) until Eq. (9) for physics-based model. In Figure 1 (a), laser absorptivity over surface temperature is plotted. Herein, the laser absorptivity of the physics-based model increases with the surface temperature and subray angle θ up to 80° . Notably, the absorptivity with a subray angle θ of 85° doesn't follow this increasing trend. It can also be observed from Figure 1 (b) that the absorptivity decreases abruptly between subrays angle $80^\circ < \theta \leq 90^\circ$. Additionally in Figure 1 (a), a discrepancy of up to 7% exists between the physics-based and empirical absorptivity with ε equal to 0.25 in the high temperature region ($T \geq 2000 \text{ K}$) and subrays angle $60^\circ \leq \theta \leq 80^\circ$. Particularly, the high temperature region is of interest, as the keyhole wall can exceed a temperature of 2000 K or above. Below the liquidus temperature, the difference is significantly higher. However, the difference of absorptivity in the low-temperature region has a little impact on the overall weld pool dynamics because the temperature in the weld pool evolves rapidly during the welding process.

Figure 1 (b) illustrates the behavior of the absorptivity w.r.t the subrays angle. It reveals that the absorptivity increases consistently with the surface temperature under a certain

incident angle of the laser beam. Furthermore, it is evident that both empirical models with ε equal to 0.3 and the literature value of 0.25 overestimate the physical accurate absorption values, while the calibrated parameter of 0.2 lies in between them.

Based on this comparison, it can be demonstrated that the commonly used empirical model accurately represents absorption behavior only if the value of ε is properly adjusted. In contrast, the implemented physics-based absorption model provides a comprehensive spectrum of absorptivity across the whole temperature range of interest and different subrays angles, self-aligning more closely with reality during the welding process without any empirical validation parameters.

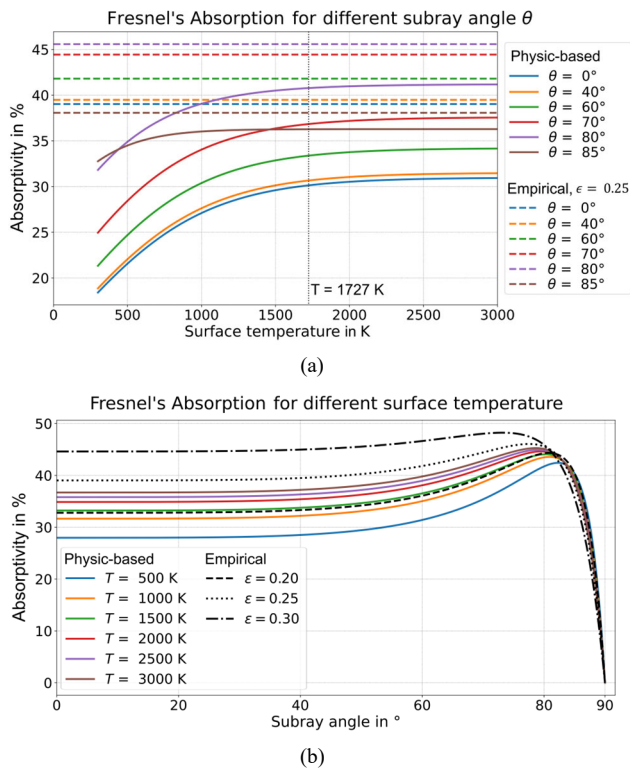


Figure 1. Plot of the absorptivity over (a) surface temperature and (b) the subray angle to the surface normal for physics-based and empirical model.

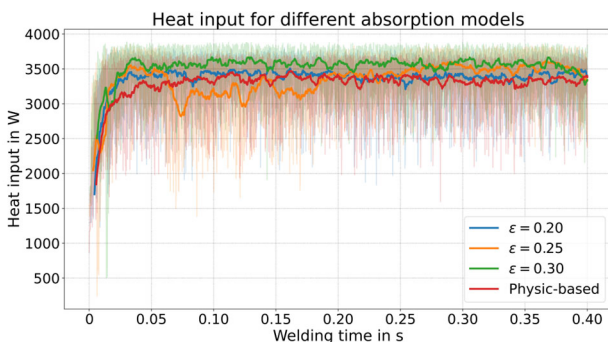


Figure 2. Plot of heat input over welding time for physics-based and empirical absorption model.

4.2. Heat input comparison

The heat input of the implemented absorption models with progressing welding time is compared in Figure 2, in reference to the works of Meng et al. [9]. To emphasize its behavior with time, the heat input is smoothed using a moving average

method. It is evident that the averaged heat input of the physics-based absorption model is slightly lower than the empirical model. Averaged over the simulated welding time up to 400 ms, the physics-based absorption model yields a thermal efficiency approximately 83% (see Figure 2), which is comparable to [12, 16].

4.3. Comparison of cross-section images

To validate the simulation results, the calculated melt contours are compared with the experimental results in Figure 3 (a). Noticeably, the physics-based absorption model with temperature dependency in Figure 3 (b) provides the best agreement with the experiment result, both in the penetration depth and melt pool width. It can be observed in Figure 3 (c) to (e) that the penetration depth increases with the calibration parameter ε . Herein, the ε value of 0.2 yields the most realistic result among all empirical absorption models. Higher calibration parameters such as 0.25 and 0.3 resulted in full penetration due to overestimating the absorptivity shown in Figure 1 (b). For the simulation cases with full penetration, it is expected that the melt pool width becomes narrower since the melt pool is not supported against gravitational pull. Nonetheless, a quantitative comparison of melt contour between simulation cases with full penetration is not necessary since the surface tension and Marangoni shear stress at the bottom plate side are not considered. With a closer look at the heat input plots in Figure 2, it can be found that the penetration depths have a big difference despite of similar averaged heat input. This means that the spatial distribution and temperature dependency of laser absorption is very important. With this comparison, the physic-based absorption model has been validated and seen as reliable method to elucidate the absorption process during LBW.

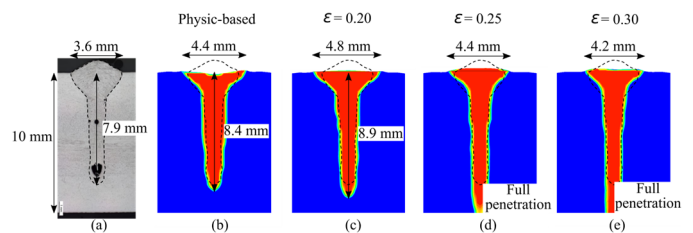


Figure 3. Comparison of the melt contour between (a) experiment result, (b) physics-based absorption model, (c) $\varepsilon = 0.2$, (d) $\varepsilon = 0.25$, and (e) $\varepsilon = 0.3$.

5. Conclusion

In this study, the empirical and physics-based absorption model have been implemented and studied. Based on the results in Chapter 4, the following conclusions can be drawn:

- The implementation of the physics-based absorption model allows for a physically accurate description of laser energy absorption in laser beam deep welding avoiding the need for a calibration, thus reducing the computational effort.
- The physics-based absorption model has yielded realistic heat input and melt pool geometry.
- By comparing the cross-section images, a successful validation for AISI 304 is achieved and the importance of temperature dependency of laser absorption is shown.

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